SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: **Total Marks:** **Duration: 90 Minutes**

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)

13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)
16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I M.GOKULAVASAN(192312477)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Article Review					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks

Introduction - The rapid growth of mobile data, 5G/6G networks, IoT devices, and connected applications has created a strong demand for efficient spectrum utilization. Traditional spectrum management relies mainly on static and exclusive allocation, which can leave spectrum underutilized. **AI-driven Dynamic Spectrum Access (DSA)** provides a more flexible approach by using Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) to dynamically identify, allocate, and share available spectrum resources.

Working Principle - In AI-driven spectrum sharing, the network continuously monitors spectrum occupancy and communication conditions. AI models analyze factors such as available channels, traffic demand, interference, and channel conditions to make appropriate spectrum allocation decisions. The system can dynamically assign spectrum to users while minimizing interference between primary and secondary users. This enables better utilization of available frequency resources while maintaining the required Quality of Service (QoS).

Role of AI and Machine Learning - AI enables spectrum management systems to make intelligent and adaptive decisions. Machine Learning and Deep Learning can be applied for spectrum sensing, occupancy prediction, traffic analysis, and interference mitigation. AI can also support real-time decision-making in changing wireless environments. Edge AI and federated learning can distribute computational workloads and help satisfy the low-latency requirements of dynamic spectrum allocation.

Major Challenges - Several technical and regulatory challenges must be addressed. **Real-time interference management** requires accurate models that can adapt to rapidly changing network conditions. AI systems also require high-quality training data, while spectrum occupancy data may be restricted because of privacy and security concerns. In addition, real-time allocation involves computational and latency constraints.

From a regulatory perspective, existing spectrum policies are primarily designed for static allocation and may not fully support automated AI-based decisions. Another challenge is the lack of transparency of some AI models. **Explainable AI (XAI)** can help regulators understand and validate spectrum management decisions. Cross-border coordination and standardization are also necessary to achieve global interoperability.

Proposed Solutions - Three important approaches are proposed for effective integration of AI-driven spectrum sharing. First, **regulatory sandboxes** can provide controlled environments for testing AI-based spectrum management before large-scale deployment. Second, **AI-enabled spectrum databases** can dynamically track spectrum occupancy and provide real-time recommendations. Third, **hybrid spectrum access models** can combine licensed, unlicensed, and dynamically allocated spectrum to improve overall spectrum utilization.

Advantages and Future Scope - AI-driven dynamic spectrum sharing can improve spectrum efficiency, reduce interference, support increasing network traffic, and enable flexible resource allocation in 5G and beyond networks. Future research can focus on developing more adaptive, robust, explainable, and low-latency AI algorithms that operate reliably within regulatory constraints.

Conclusion - AI-driven Dynamic Spectrum Sharing represents a major shift from traditional static spectrum management toward intelligent and adaptive wireless resource utilization. By combining AI-based spectrum sensing, prediction, and allocation with suitable regulatory frameworks, 5G and future 6G networks can achieve more efficient spectrum sharing while maintaining interference control and QoS. Regulatory sandboxes, AI-enhanced spectrum databases, hybrid access models, explainable AI, and international standardization can play important roles in making this technology practical and reliable.

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AI-Powered Dynamic Spectrum Sharing In 5G Networks: A Federated Learning Approach

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***Abstract*—Dynamic Spectrum Sharing is a critical technology in 5G networks, enabling real-time allocation of spectrum resources based on traffic demand, interference, and network conditions. Traditional spectrum management approaches, often centralized and static, struggle to adapt to the heterogeneous and rapidly changing environments of modern wireless networks. Moreover, they involve high communication overhead and pose risks to data privacy. To address these limitations, this study proposes an Artificial Intelligence-powered dynamic spectral sharing framework utilizing Federated Learning, where fifth-generation (5G) base stations collaboratively train a global model without sharing raw data. Each base station performs local training based on real-time spectrum usage and transmits only model updates to a central aggregator, which combines them using the Federated Averaging algorithm. This decentralized approach preserves privacy, reduces latency, and ensures scalable, region-aware spectrum optimization. Simulation results demonstrate significant improvements in throughput, reduced latency, and enhanced spectrum efficiency compared to traditional methods. The model achieves faster convergence and adapts well to diverse environments, proving the effectiveness of FL in dynamic spectrum sharing. The proposed approach establishes a solid basis for intelligent, privacy-preserving, and scalable spectrum management in emerging network architectures.**

Keywords—Dynamic Spectrum Sharing, Artificial Intelligence, Federated Learning, Fifth-Generation, Base Station

I. INTRODUCTION

The evolution of wireless communication technologies has led to an exponential increase in the demand for spectrum resources. With the rollout of 5G networks, the pressure on the radio frequency spectrum has intensified due to the growing number of connected devices, bandwidth-intensive applications, and diverse service requirements. Traditional spectrum management techniques, which rely on static or semi-static allocation policies, are increasingly inadequate in addressing the dynamic and context-sensitive nature of modern wireless communication [1]. Dynamic Spectrum Sharing (DSS) addresses this challenge by allowing spectrum bands to be allocated and reused adaptively, based on real-time parameters such as traffic load, interference, and user mobility [2]. DSS improves spectral efficiency by ensuring that underutilized spectrum bands are accessed opportunistically by other users or systems, thus reducing the problem of spectrum scarcity [3]. Fig. 1 visualizes the DSS. To intelligently manage spectrum in real-time and under complex conditions, Artificial Intelligence (AI) and Machine Learning (ML) algorithms have been introduced into DSS frameworks [4]. AI enables base stations to analyze real-time data and make predictions or decisions about spectrum allocation, such as selecting optimal frequency bands or adjusting transmission power. However, the integration of centralized AI systems into DSS introduces several practical and privacy-related limitations.

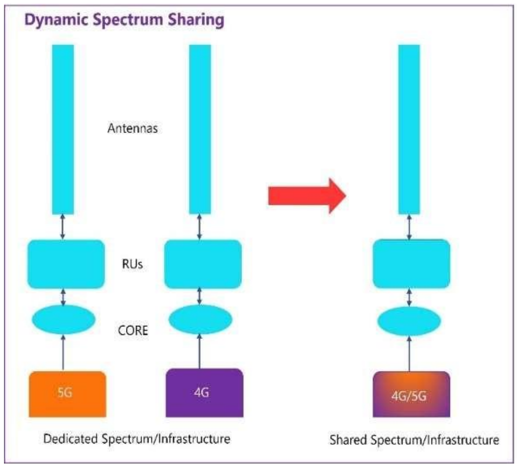

Fig. 1. Dynamic Spectrum sharing

Centralized models require large volumes of labeled data to be transmitted from distributed nodes to a central server, which increases communication overhead, creates latency bottlenecks, and raises serious data privacy concerns, particularly in telecom systems that manage sensitive user traffic. Moreover, centralized approaches lack adaptability in highly distributed and heterogeneous 5G environments where conditions vary widely across locations. Federated Learning (FL) has emerged as an effective alternative that resolves these limitations by decentralizing the training process [5]. In an FL framework, each base station trains its model locally using data collected from its specific spectrum environment and user traffic, as shown in Fig. 2. Rather than sending raw data to a central server, each Base Station (BS) transmits only its model updates, for example, trained weights or gradients. These updates are aggregated at a central node referred to as the central aggregator, using techniques like Federated Averaging (FedAvg). This aggregation results in a global model that benefits from the collective experience of all participating BSs without ever exposing their local datasets. The global model is then redistributed to the BSs, which continue local training using their updated data. This iterative learning process continues across several communication rounds, refining the model progressively [6].

In a federated architecture, each 5G BS trains locally and makes spectrum decisions for connected user equipment using the latest global model [7]. Local models are uploaded to a central aggregator, which combines and redistributes them without exposing raw data. This approach preserves privacy, reduces latency, and supports scalability across diverse environments. By enabling intelligent, region-aware spectrum allocation, the federated DSS model ensures secure, real-time

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optimization. It is especially suited for multi-operator networks, critical IoT, and public safety, positioning FL as vital for future wireless systems [8].

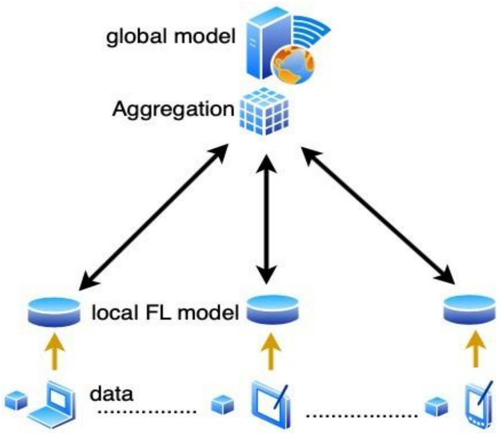

Fig. 2. Basic Architecture of Federated learning

This study presents an AI-powered framework for DSS in 5G networks using FL. By enabling distributed base stations to collaboratively learn spectrum access strategies without sharing raw data, the proposed model enhances spectral efficiency, reduces interference, and ensures privacy preservation. The framework addresses key limitations of centralized learning approaches, such as data privacy concerns, high communication overhead, and lack of adaptability in heterogeneous environments. Through iterative local training and global model aggregation, the system evolves into a robust and scalable spectrum management solution, tailored for real-time deployment in next-generation wireless networks. The objectives of the proposed study are:

- To design a privacy-preserving, AI-driven framework for DSS sharing in 5G, where BS collaboratively learn optimal allocation strategies using FL without exchanging raw data.
- To enhance spectral efficiency, reduce interference, and support real-time, decentralized decision-making, by enabling intelligent local model training and secure global model aggregation across distributed base stations.

II. RELATED WORKS

Wang et al. [9] proposed a DSS framework for smart grids and power communication networks during emergencies, leveraging Deep Reinforcement Learning (DRL) and FL. The approach employed the Maximum Entropy based Multi-Agent Actor-Critic (ME-MAAC) algorithm as a local learning model, enabling agents to train actor and critic networks independently. Simulation results demonstrated improved performance in terms of accumulated reward, access rate, and convergence speed compared to traditional methods. However, a key limitation identified was the insufficient research on security measures. Biswas et al. [10] developed a full-duplex (FD) enabled Integrated Access and Backhaul (IAB) networks operating in millimeter-Wave frequencies addressed the challenge of self-interference using a subarray-based hybrid precoding scheme. Zero-forcing and Minimum Mean Square Error (MMSE) methods were employed to mitigate multiuser interference and residual SI, respectively. Simulation results demonstrated that, despite radio frequency insertion loss, nearly double Spectral Efficiency (SE) was achieved compared to half-duplex systems under low channel estimation error (CEE). However, limitations included sensitivity to CEE, the lack of accurate SI channel estimation, and the assumption of equal power allocation, which restricts adaptability under real-world conditions with hardware constraints.

Song et al. [11] introduced an FL-based framework for dynamic spectrum access (DSA), addressing limitations of traditional model-based approaches in under-measured environments. The model utilized Multi-Agent Reinforcement Learning (MARL) to approximate network dynamics while preserving terminal privacy under heterogeneous data. Simulation results showed that the FL-enabled MARL approach improved spectrum efficiency and system throughput using policy gradient algorithms, outperforming conventional learning methods. However, limitations included challenges in managing heterogeneous data distributions, ensuring stable convergence across agents, and the complexity of coordination in distributed environments. Liu et al. [12] addressed multi-dimensional resource allocation in cognitive radio systems using DRL, where secondary users (SUs) were modelled as agents making channel and power control decisions. The proposed approach utilized Deep Q-Network (DQN) and Deep Recurrent Q-Network (DRQN) architectures, incorporating techniques such as experience replay and target network freezing. Simulation results showed that the method significantly improved user rewards and reduced collisions, outperforming the opportunistic multichannel by approximately 10% in single SU and 30% in multi-SU scenarios. However, the approach faced limitations related to computational complexity and sensitivity to parameter settings, requiring careful tuning for optimal performance.

Vo et al. [13] proposed an FL-enabled spectrum sharing framework for 6G networks that emphasizes security and privacy preservation during collaborative training. The method incorporated adversarial robustness techniques to mitigate AI-powered threats such as adversarial jammers and compromised secondary users. The model analyzed threat models specific to 6G environments and demonstrated, through simulations, that traditional FL defenses were inadequate in wireless scenarios due to dynamic environments and distributed user behavior. The analysis highlighted vulnerabilities in model aggregation and the risks of gradient leakage. While the framework improved resilience compared to baseline FL models, it faced limitations in scalability and computational overhead, especially when employing encryption techniques. A consortium blockchain-based DSS framework was proposed by Li [14] to ensure secure, efficient spectrum allocation with guaranteed revenue and Quality of Service (QoS). The framework allowed mobile network operators to act as either spectrum providers or requestors, with all transactions managed via smart contracts on Hyperledger Fabric. A multi-leader multi-follower Stackelberg game model was used to derive optimal pricing and purchasing strategies, solved using the RMSProp algorithm. Simulation and prototype results demonstrated the framework’s feasibility, showing that spectrum allocation fairness and transaction transparency were achieved. However, as the number of participants increased, both submission time and average latency also increased, indicating scalability challenges and potential performance bottlenecks in large-scale deployments.

Wang et al. [15] proposed a Smoothing Deep Q-Network (SDQN) to enhance spectrum sharing efficiency in cognitive radio systems, where secondary users access the spectrum without interfering with primary users. The SDQN integrated a kernel-based adaptive filter to smooth Q-value outputs and employed a historical weighting mechanism to mitigate over-smoothing. Simulation results demonstrated that SDQN outperformed conventional DQN by producing more stable and accurate power control strategies, effectively improving spectral efficiency and minimizing interference in dynamic radio environments. This highlights SDQN's potential for robust learning in noisy wireless settings. A blockchain-based dynamic spectrum sharing framework was proposed by Zhu et al. [16] to ensure decentralization, transparency, and privacy in untrusted environments. A privacy-preserving double auction mechanism was developed using differential privacy to protect user data and encourage participation. The winner determination problem was solved using a DRL approach integrated into the blockchain consensus process. Simulation results demonstrated superior performance over traditional methods in terms of spectrum allocation efficiency and privacy preservation. However, while the system ensured truthfulness and individual rationality, limitations included computational complexity and reliance on blockchain latency and DRL stability, which affect scalability in real-time applications.

Chang et al. [17] proposed a Fed-MADRL framework combined FL with Multi-Agent Deep Reinforcement Learning (MADRL) to design a collaborative dynamic spectrum access strategy. FL allowed users to optimize a shared policy without exposing local data, preserving privacy and improving communication efficiency by sharing only quantized model updates. This approach enabled distributed learning without requiring global state information. Simulation results showed that Fed-MADRL significantly outperformed independent learning methods and achieved performance close to centralized MADRL approaches, but with reduced communication costs. A limitation was the loss in learning precision due to quantization, which could affect model accuracy in more complex network scenarios. A hierarchical game-theoretic model was proposed by Lim et al. [18] to model edge association and resource allocation in Hierarchical Federated Learning (HFL) networks. The method employed an evolutionary game to capture worker-edge server association dynamics and a Stackelberg differential game to optimize reward and bandwidth allocation by model owners and edge servers. Numerical results validated the model’s effectiveness in capturing system dynamics under network heterogeneity. The framework demonstrated strong adaptability and coordination in self-organizing HFL environments. However, a limitation was the focus on a single model owner.

Despite the growing interest in applying FL to DSS in 5G networks, several key limitations persist that highlight critical research gaps. Existing studies often lack robust security and privacy-preserving mechanisms, particularly in protecting against model inversion attacks and adversarial behavior during aggregation. Many approaches assume ideal conditions, such as equal power allocation and accurate channel estimation, which limit adaptability in real-world hardware-constrained environments [10]. Moreover, handling heterogeneous data distributions, ensuring convergence across distributed agents, and maintaining coordination at scale remain complex and underexplored challenges. Computational overhead, sensitivity to parameter settings, and the scalability of secure protocols like blockchain further constrain practical deployment [12, 13]. Additionally, precision loss due to model quantization and limited consideration of multi-model ownership scenarios indicate the need for advanced techniques such as adaptive encryption, auction-based resource pricing, and resilient aggregation strategies. These gaps underscore the necessity for a more secure, scalable, and context-aware FL framework for AI-powered DSS in 5G networks.

III. METHODOLOGY

The workflow of AI-powered dynamic spectrum sharing using FL involves several key steps, as shown in Fig. 3. First, user equipment transmit data to nearby 5G base stations (BSs), which collect local observations such as traffic demand, interference levels, and channel occupancy. Each BS then trains a local ML model on its own data without sharing raw information. The locally trained model parameters are securely uploaded to a central aggregator, which performs federated averaging to create a global model. This updated global model is distributed back to the BSs, enabling them to make intelligent, real-time spectrum allocation decisions. This decentralized approach ensures privacy, reduces communication overhead, and improves adaptability across diverse network environments.

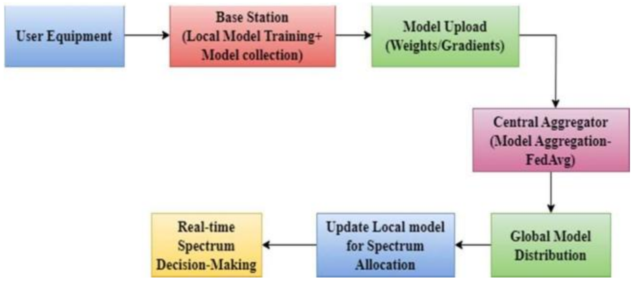

Fig. 3. Block diagram of proposed model

A. System Architecture Overview

The proposed system architecture for DSS in 5G networks is built on the principles of FL, designed to enable decentralized intelligence while preserving user privacy. At the core of this architecture lies a central parameter server, which is responsible for aggregating local model updates, maintaining the global model, and coordinating training across a network of distributed clients. These clients are 5G BSs, each equipped with the computational capability to locally train ML models using real-time data gathered from their respective environments. Each BS functions as an autonomous federated client, independently collecting and processing localized data. The data captured by the BS includes a range of features essential for efficient spectrum management, such as channel occupancy, user traffic demand, interference levels, and signal-to-noise ratios (SNRs). This data reflects the dynamic nature of the local wireless environment and forms the basis for learning optimal spectrum allocation strategies. Importantly, raw data is never transmitted outside the BS, ensuring data privacy and compliance with security standards. Instead, only the locally trained model parameters are shared with the central server. Fig. 4 illustrates the architecture of AI-powered DSS in 5G networks using FL.

Communication between the BS and the central server occurs over secure, low-latency links, typically facilitated through the 5G core network infrastructure. These communication rounds are periodic and lightweight, designed to minimize bandwidth consumption while ensuring timely synchronization of the global model. The use of encrypted transmission protocols further secures the exchange of model updates, preventing eavesdropping or tampering during transmission.

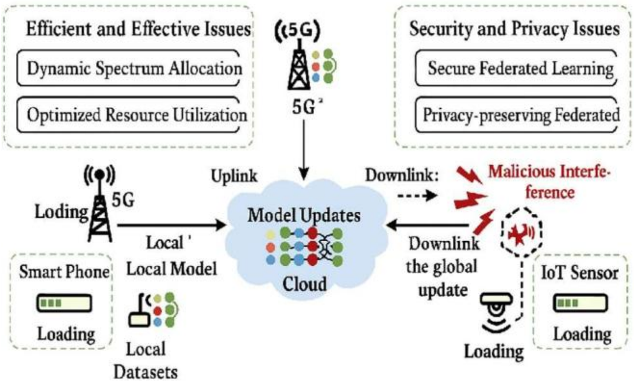

Fig. 4. Architecture of AI-Powered DSS in 5G Networks Using FL

This distributed and privacy-preserving architecture offers several advantages. By decentralizing the learning process, the system becomes highly scalable, able to accommodate a large number of BSs spread across a wide geographical area. It also enhances resilience and adaptability, as each BS tailors its model to the unique characteristics of its local environment while still benefiting from the shared global knowledge. Furthermore, by limiting data movement and performing computation at the edge, the system significantly reduces communication overhead and ensures real-time responsiveness, both of which are critical for dynamic spectrum sharing in ultra-dense 5G deployments.

B. Data Collection at Local Base Stations

In the proposed FL framework for dynamic spectrum sharing in 5G, data collection at the BS level plays a critical role in enabling intelligent, context-aware decision-making. Each 5G BS, operating as an independent FL client, continuously monitors its local radio environment and user behavior to gather essential time-varying metrics. These metrics provide a real-time snapshot of the network's operational state and are used as input features for local model training. One of the primary metrics collected is the spectrum occupancy $O_k(t)$, which indicates the status of each frequency channel, whether it is currently free or occupied. This information is vital for identifying available spectrum resources and avoiding interference with ongoing transmissions. Simultaneously, each BS monitors the user demand per channel $D_k(t)$, representing the number of active users or data sessions requesting access to specific channels. This helps in determining traffic load and planning for efficient allocation.

Another critical metric is interference level $I_k(t)$, which reflects the impact of adjacent BSs or primary users on the QoS within the cell. High interference leads to degraded performance and is therefore a key factor in the optimization process. Additionally, traffic type and QoS requirements $T_k(t)$ are logged to understand the nature of the applications being served such as voice, video, or mission-critical IoT traffic and their specific latency, bandwidth, and reliability constraints. All these parameters are collectively stored in the local dataset D_k, for the k[th] BS. Crucially, this data is never uploaded or shared with external entities or central servers, thereby preserving user privacy and complying with data protection regulations. Instead, the data is used locally to train an ML model that captures patterns in spectrum usage and learns optimal allocation strategies under the current network conditions. By leveraging rich, real-time environmental and traffic information, each BS can contribute to the global learning process in a privacy-aware and context-sensitive manner. This localized data-driven intelligence is what enables FL to outperform traditional centralized approaches in dynamic, heterogeneous 5G network scenarios.

C. Local Model Training at Base Stations

After collecting localized data on spectrum usage, user demand, interference levels, and traffic types, each BS proceeds to train a ML model locally. This model aims to learn an optimal policy for dynamic spectrum allocation by analyzing the patterns and relationships in the data over time. This training happens entirely on-device, ensuring that sensitive network and user data remains private and secure. The local dataset at the kth BS is denoted by D_k, and expressed as in (1).

$$D_k = \{(x_i^k, y_i^k)\}_{i=1}^{n_k} \quad (1)$$

where x_i^k is the feature vector and y_i^k is the label or target representing the optimal spectrum action (e.g., channel and power allocation). The feature vector includes Spectrum occupancy history, Interference levels, Traffic demand and QoS-related metrics. Each BS trains a model $f(w_k;\ x_i^k)$ parameterized by w_k, by minimizing a local empirical loss function L_k over its dataset as in (2).

$$L_k(w_k) = \frac{1}{n_k}\sum_{i=1}^{nk} l(f(w_k; x_i^k), y_i^k) \quad (2)$$

where $l(\cdot,\cdot)$ is the loss function, typically Mean Squared Error (MSE) for regression tasks or cross-entropy loss for classification. Training is performed using Stochastic Gradient Descent (SGD) or an adaptive optimizer like Adam, with updates of the form, as in (3).

$$w_k^{(t+1)} = w_k^{(t)} - 5.\ AL_k(w_k^{(t)}) \quad (3)$$

where 5 is the learning rate, t is the local training iteration, $AL_k(w_k^{(t)})$ is the gradient of the loss with respect to the model parameters. The gradient of the local loss at BS k is denoted as g_k, expressed as in (4).

$$g_k = AL_k(w_k^{(t)}) \quad (4)$$

Then, the model is updated via stochastic gradient descent (SGD) as in (5).

$$w_k^{(t+1)} = w_k^{(t)} - 5.\ g_k^{(t)} \quad (5)$$

or, using Adam optimizer as in (6).

$$w_k^{(t+1)} = w_k^{(t)} - 5.\ \frac{\hat{m}_k^{(t)}}{\sqrt{\hat{v}_k^{(t)}} + \epsilon} \quad (6)$$

where $\hat{m}_k^{(t)}$ bias-corrected first moment estimate, $\hat{v}_k^{(t)}$ bias-corrected second raw moment estimate and ϵ is the small constant to prevent division by zero. Each BS performs this update over multiple local epochs before participating in the global aggregation step.

The training continues for several local epochs, enabling each BS to fine-tune its model based on local traffic conditions and interference patterns. This allows the model to learn nuanced strategies, such as which channels to avoid during high-interference periods or how to prioritize different user QoS requirements when spectrum is scarce. To prevent overfitting and stabilize training, techniques like dropout, batch normalization, or early stopping are applied. Additionally, each BS caches past experiences or traffic logs to train on varied scenarios, improving generalization. After local training is complete, the BS does not send its raw data but only shares the updated model weights w_k with the central server for federated aggregation, preserving privacy.

D. The Role of Federated Learning in the Proposed Model

Once local training is completed at each BS, only the model parameters w_k are sent to a central server, preserving data privacy. Unlike traditional centralized learning, where raw data is uploaded for processing, FL ensures that sensitive information such as spectrum usage and user traffic remain on the local device. This approach minimizes communication overhead and enhances security, making it more suitable for large-scale, privacy-sensitive 5G deployments [19]. FL aligns with modern data protection regulations like General Data Protection Regulation, addressing the risks associated with centralized data handling.

E. Model Aggregation via Federated Averaging (FedAvg)

Model aggregation in FL involves combining locally trained models from distributed clients, such as 5G BS, into a single global model without sharing raw data. Each client sends only model parameters to a central server, which uses the FedAvg algorithm to compute a weighted average based on dataset size. This approach preserves privacy, reduces communication overhead, and enables collaborative learning across the network, making it well-suited for DSS in 5G. The central server aggregates the local models using the FedAvg algorithm, which computes a weighted average of all received model parameters, as in (7).

$$w^{(t+1)} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{(t+1)} \tag{7}$$

where $w^{(t+1)}$ is the global model after aggregation, n_k is the number of training samples at BS k, $n = \sum_{k=1}^{K} n_k$ is the total number of samples across all participating BSs. This weighted aggregation ensures that BS with more representative data have a proportionally greater influence on the global model.

F. Model Distribution and Iterative Training

Following the model aggregation phase in FL, the newly computed global model is broadcast back to all participating clients, which, in the case of DSS in 5G networks, are distributed base stations. This process is known as model distribution. The central server disseminates the updated global model parameters $w^{(t+1)}$ to each BS. These parameters now represent a generalized understanding of spectrum allocation strategies learned from the diverse and distributed local training data of all BSs in the system. Upon receiving the global model, each BS updates its local model to match the aggregated one and resumes training on its most recent or continuously updated local dataset. This begins the next round of FL, where each BS further refines the shared model using its unique spectrum observations, traffic demands, and interference conditions. The local training process again involves optimizing the objective function as in (8).

$$L_k(w_k) = \frac{1}{n_k} \sum_{i=1}^{nk} l(f(w_k; x_i^k), y_i^k) \tag{8}$$

This iterative cycle of local training, model uploading, aggregation, and redistribution continues across several communication rounds. The goal is to progressively refine the global model until it converges to an optimal or stable state. Each iteration allows the system to adapt to dynamic and time-varying wireless environments, which is particularly valuable in 5G networks where user density, interference patterns, and service requirements fluctuate frequently. By executing inference locally with the updated model, each base station achieves low-latency, real-time spectrum access decisions, while simultaneously benefiting from the shared insights of global learning. Over time, as more data is observed and incorporated into local training, and as more communication rounds are completed, the global model becomes increasingly robust, capable of generalizing across heterogeneous radio environments. The iterative training also supports continual learning, enabling the model to evolve alongside changing network topologies and usage behaviors. The model distribution and iterative training phases ensure that FL in DSS systems remains adaptive, decentralized, and scalable. This cyclical learning process not only facilitates efficient spectrum management but also preserves data privacy, reduces communication overhead, and supports deployment across a wide range of network conditions in future 5G and beyond-5G systems.

G. Security and Privacy Preservation in Federated DSS

One of the fundamental motivations behind adopting FL for DSS in 5G networks is its inherent capability to preserve privacy and enhance security [20]. Traditional centralized ML systems require raw data to be transferred from local devices to a central server, which poses significant risks in terms of data leakage, user privacy violations, and non-compliance with regional data governance regulations. In contrast, FL shifts the paradigm by allowing each BS to train models locally on its own private dataset and share only the learned model parameters such as weights or gradients with a central aggregator. This privacy-by-design approach significantly reduces the attack surface and data exposure inherent in centralized architectures.

1) Data Locality and Raw Data Protection

The primary privacy-preserving aspect of FL is that raw data never leaves the BS. Each BS keeps its data, comprising real-time spectrum usage, interference metrics, and user-specific traffic patterns, strictly local. Instead of transmitting this potentially sensitive data to a remote server, only abstract representations in the form of model updates are shared. This design eliminates the risk of raw data interception or misuse during transmission and ensures compliance with data protection laws, such as GDPR, HIPAA, or regional telecom-specific policies.

2) Differential Privacy

To strengthen the privacy guarantees further, Differential Privacy (DP) is incorporated into the FL workflow. Differential privacy ensures that the output of a learning algorithm does not reveal whether any single data point was part of the training dataset [21]. In practice, this is achieved by adding calibrated random noise to the model updates before they are shared, as in (9).

$$\widetilde{w}_k = w_k + N(0, \sigma^2) \tag{9}$$

where $N(0, \sigma^2)$ is Gaussian noise with variance σ^2 , and $\widetilde{w}_k$ is the noisy version of the model update. This technique ensures that even if an attacker gains access to the transmitted model weights, they cannot reliably infer any information about individual users or spectrum events at the BS level.

3) Secure Aggregation

Another key privacy-enhancing mechanism is secure aggregation, which ensures that the central server computes the aggregated global model without being able to view any individual BS's model update. This is typically achieved using cryptographic techniques such as homomorphic encryption or secure multiparty computation. In secure aggregation, each BS encrypts its model update before transmission, and the server only sees the encrypted sum, as in (10).

$$\text{Server receives: } \sum_{k=1}^{K} Enc(w_k) \rightarrow Dec(\sum_{k=1}^{K} w_k) \quad (10)$$

where $Enc(\cdot)$ and $Dec(\cdot)$ denote encryption and decryption operations, respectively. This approach protects each BS's individual contribution, even from the central server itself, thereby enhancing trust in federated deployments across multiple organizations or operators.

4) Authentication and Authorization

To prevent unauthorized devices or malicious actors from participating in the FL process, strong authentication and authorization mechanisms are essential. Public-key infrastructure, digital signatures, or hardware-based trusted execution environments are used to verify the identity of each BS and ensure that only legitimate nodes upload and receive model parameters.

5) Robustness Against Adversarial Attacks

Federated DSS systems are resilient to poisoning attacks, where a malicious BS submits manipulated model updates to degrade the global model's performance [22]. Techniques such as anomaly detection, Byzantine-resilient aggregation rules, and reputation scoring systems are employed to identify and isolate malicious participants without disrupting the overall learning process.

H. Hardware and Software

The implementation of an FL-based DSS framework in 5G networks requires a cohesive hardware and software setup. Each 5G BS acts as a federated client, equipped with multicore CPUs and GPUs to support real-time data processing and local model training. These edge devices use Software-Defined Radios (SDRs) to monitor spectrum usage, interference, and signal quality in real-time. Reliable, low-latency communication interfaces connect the base stations to a central aggregator server, which coordinates the FL process. The central server aggregates model updates from all participating base stations using the FedAvg algorithm and distributes the updated global model back to each client. It operates from an edge data centre with sufficient computational ability to handle concurrent communications and model aggregation tasks securely and efficiently. On the software side, FL workflows are managed using frameworks such as TensorFlow Federated, while ML models are developed using TensorFlow or PyTorch. MATLAB is used to simulate realistic 5G environments and validate system performance. Lightweight communication protocols like google Remote Procedure Call (gRPC) manage data exchange, and Docker is used for scalable deployment. To ensure privacy and security, differential privacy and secure aggregation techniques are implemented, preserving data confidentiality while enabling collaborative learning across the network.

IV. RESULTS AND DISCUSSION

In wireless networks, latency, throughput, and spectrum efficiency are key performance indicators used to evaluate the effectiveness of spectrum sharing strategies. Latency (L) refers to the time delay between a user request and the network response, typically measured in milliseconds (ms), and is minimized to ensure real-time communication, as in (11).

$$L = T_{\text{processing}} + T_{\text{queue}} + T_{\text{transmission}} T_{\text{propagation}} \quad (11)$$

$T_{\text{processing}}$ is the time to process the request at the BS, T_{queue} is the time spent waiting in the transmission queue, $T_{\text{transmission}}$ is the time to transmit the packet and $T_{\text{propagation}}$ is the time taken for the signal to travel over the channel. Fig. 5 illustrates the reduction in latency over successive federated learning rounds in a wireless network scenario. Initially, at round 1, the latency is approximately 24.5 ms. As the number of rounds increases, latency steadily declines dropping to around 22 ms at round 2, 19 ms at round 3, 17 ms at round 4, and reaching 15 ms at round 5. This consistent downward trend demonstrates the effectiveness of the federated learning approach, where the global model improves through collaboration without sharing raw data. As a result, base stations are able to make faster and more accurate spectrum access decisions locally, reducing communication delays. The observed latency reduction highlights the benefits of localized inference with globally trained models, supporting low-latency, real-time responsiveness in 5G wireless networks.

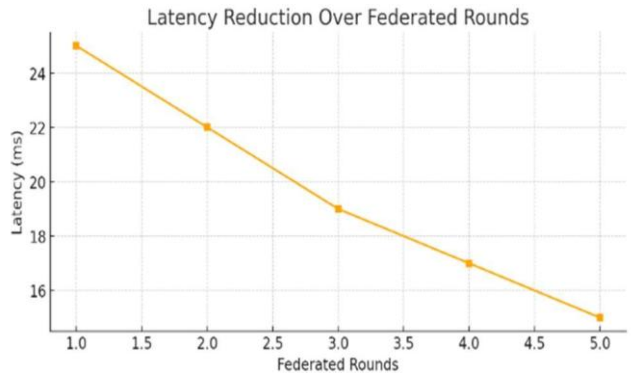

Fig. 5. Latency of the proposed model

Throughput (T) represents the amount of successful data transmission over the network, measured in megabits per second (Mbps), and is calculated by dividing the total data received by the time duration of transmission., as in (12).

$$T = \frac{D_{\text{total}}}{t} \quad (12)$$

where D_{total} is the total amount of data successfully transmitted (in bits or bytes) and t is the time duration of transmission (in seconds). System throughput is illustrated in Fig. 6. Throughput steadily increases from 50 Mbps at round 1 to approximately 72 Mbps by round 5, demonstrating a consistent improvement in data transmission efficiency as the federated learning process progresses. This upward trend reflects the model's growing ability to make more intelligent spectrum allocation decisions, minimizing contention and maximizing the use of available bandwidth. As each base station contributes to the global model while retaining local insights, the network becomes better optimized for real-time traffic demands. The increase in throughput highlights the effectiveness of the proposed framework in achieving high-performance communication, crucial for supporting bandwidth-intensive applications in 5G wireless networks.

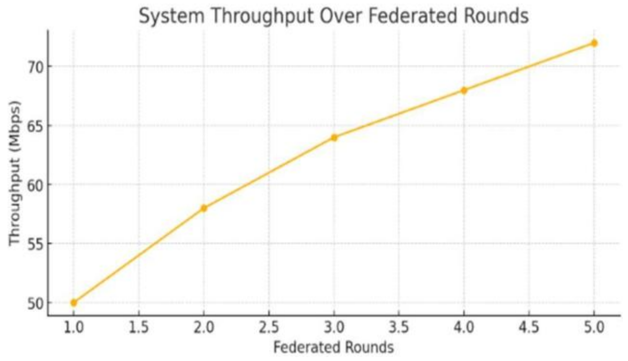

Fig. 6. Throughput of the proposed model

Spectrum efficiency 5, measured in bits per second per hertz (bits/s/Hz), reflects how effectively the available spectrum is used and is computed by dividing the total throughput by the bandwidth utilized, as in (13).

$$5 = \frac{T}{B} \tag{13}$$

where B is the bandwidth used in Hz. In the proposed model, these metrics improve over federated rounds as the global model becomes better at making dynamic, localized spectrum allocation decisions. Fig. 7 illustrates the improvement in spectrum efficiency across five federated learning rounds. Initially, at round 1, spectrum efficiency is approximately 2.3 bits/s/Hz, which progressively increases to about 3.6 bits/s/Hz by round 5. This upward trend signifies the model's enhanced ability to allocate spectrum dynamically and intelligently. As the global model is refined through federated learning, base stations are empowered to make more precise, context-aware spectrum decisions. Consequently, the network utilizes the available spectrum more effectively, achieving higher data rates without requiring additional bandwidth. This demonstrates the proposed model's potential to significantly boost the performance of 5G wireless networks in dense and dynamic environments.

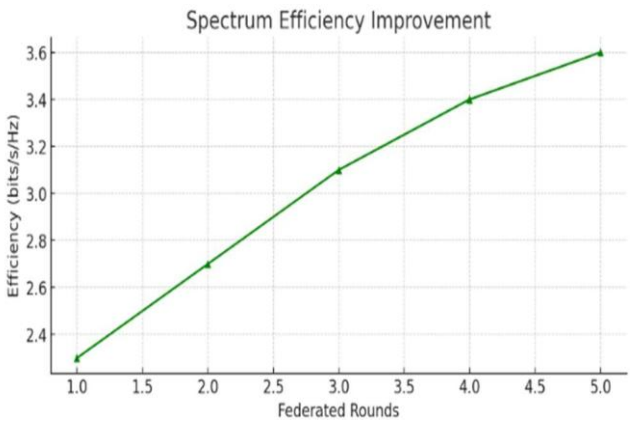

Fig. 7. Spectrum efficiency of the proposed model

The proposed FL-based spectrum sharing model demonstrates clear performance improvements across key metrics over successive training rounds. Table I shows the performance evaluation of the proposed model.

TABLE I PERFORMANCE EVALUATION OF THE PROPOSED MODEL

Federated Round	Throughput (Mbps)	Latency (ms)	Spectrum Efficiency
1	50	25	2.3
2	58	22	2.7
3	64	19	3.1
4	68	17	3.4
5	72	15	3.6

V. CONCLUSION

Effective spectrum management is essential to meet the growing demands of 5G wireless networks, where dynamic and efficient allocation of bandwidth is critical for performance. This study proposed an AI-powered dynamic spectrum sharing framework using FL, allowing distributed base stations to collaboratively train a global model without sharing raw data. The approach overcomes the limitations of traditional centralized systems by enabling real-time, privacy-preserving, and adaptive decision-making directly at the network edge. The proposed model demonstrated significant improvements in spectrum efficiency, reduced latency, and higher throughput through collaborative learning and localized optimization. By using the Federated Averaging algorithm, the system achieves stable convergence while preserving data confidentiality and minimizing communication overhead. This framework not only supports scalable deployment across heterogeneous 5G environments but also ensures secure, decentralized operation aligned with modern privacy standards.

One of the limitations of the proposed framework is that it assumes all base stations have sufficient computational resources for local model training, which may not be practical in resource-constrained or energy-limited environments. In real-world deployments, network instability, varying computational capabilities of base stations, and asynchronous updates impact the convergence speed and model accuracy. Future work will need to address these practical challenges by developing more robust, asynchronous FL techniques that can tolerate heterogeneous network conditions. By integrating lightweight encryption and adaptive model compression techniques further reduce overhead and improve scalability, paving the way for more robust, real-time spectrum sharing solutions in next-generation 6G networks.

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